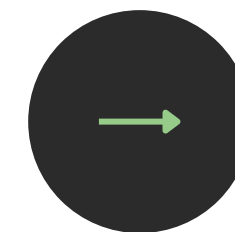


Credit Card Fraud Detector

By: Nisarga Kadam and Kevin Zhang



Credit scores impact your ability to get loans, rent, or even get hired



Agenda



Introduction

Related Works

Deep Learning Methods

Dataset & Pre-processing

Model Architecture

Model Training

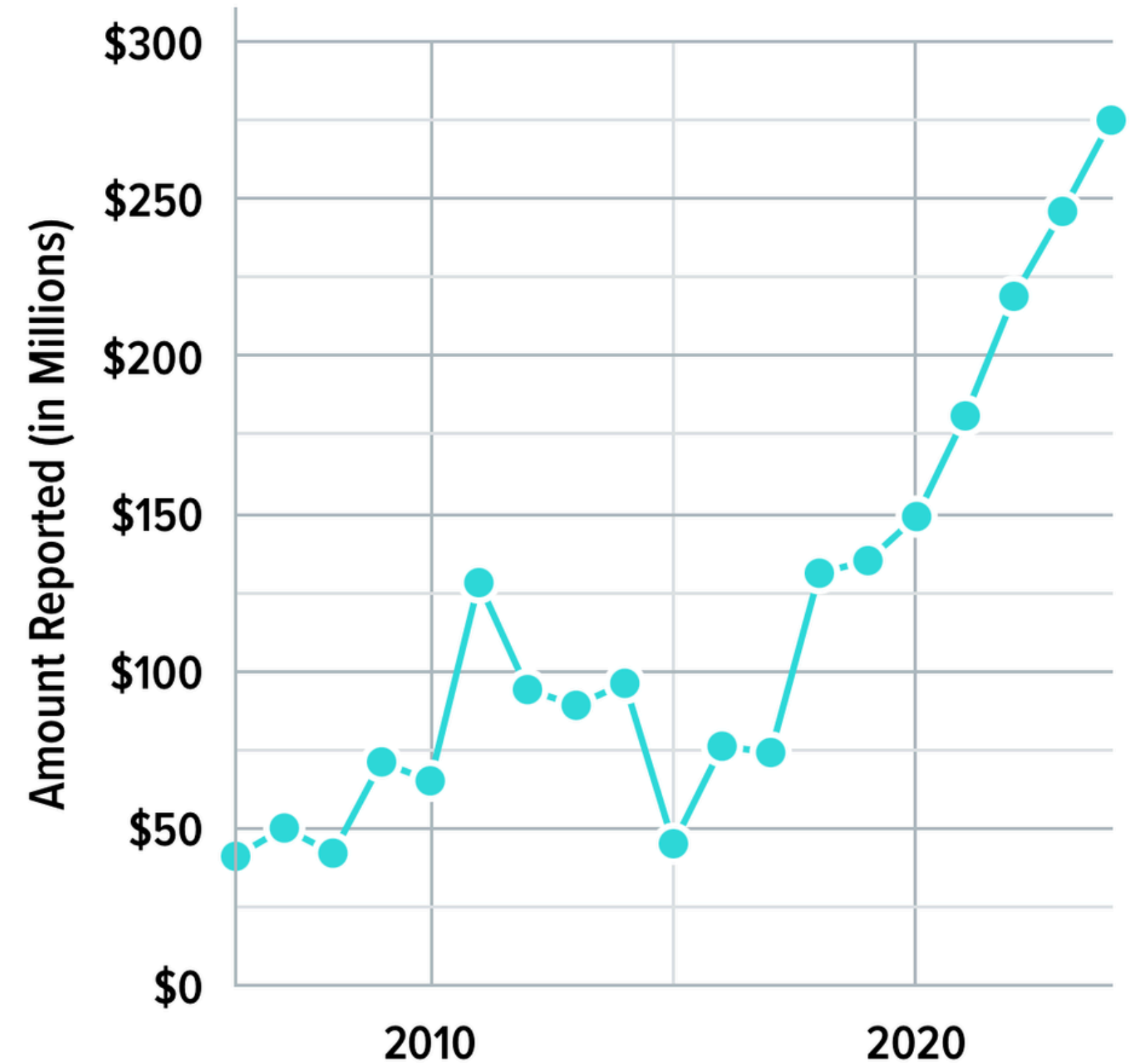
Result & Evaluation

Next Steps



Introduction

- Credit card losses were reported to be \$275 million in 2024
- Detrimental to both consumers and financial institutions
- Choosing Deep Learning methods



(Source: Consumer Sentinel Network, Annual Reports)

Related Works

It Is Not Only A Machine Learning Approach

Machine learning algorithms are not the only approach, even though they have been very popular in research nowadays.



Statistical approaches have been done and can also be viable in detecting fraud:

- Descriptive Statistics
 - utilizes mean, SD, and percentiles
- Hypothesis Testing
 - uses null and alternative hypotheses and testing (t-tests or chi-square tests)

Related Works

Techniques To Improve Performance

Besides just changing ML algorithms or using a hybrid of them, there are techniques that can be implemented prior to model training.



Due to the nature of credit card transactions, there are going to be an imbalance between legit and fraudulent transactions.

Techniques like **SMOTE** (Support Over-sampling Technique) can help the model's performance.

In one instance, a model was able to get a better accuracy by 2 to 4 percent.

Dataset & Preprocessing

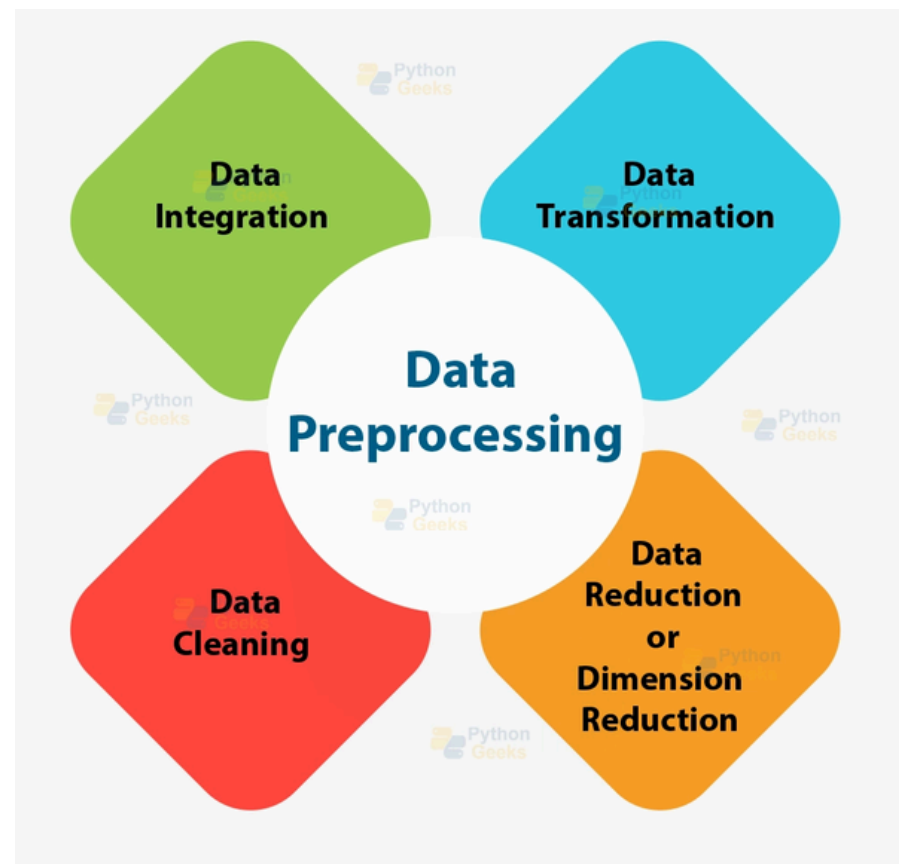
Datasets Are Not Always Readily Available

There are some datasets that are just not publicly available due to reasons of privacy and security. As a result, researchers oftentimes would need to seek alternative ways to get these data to compensate.

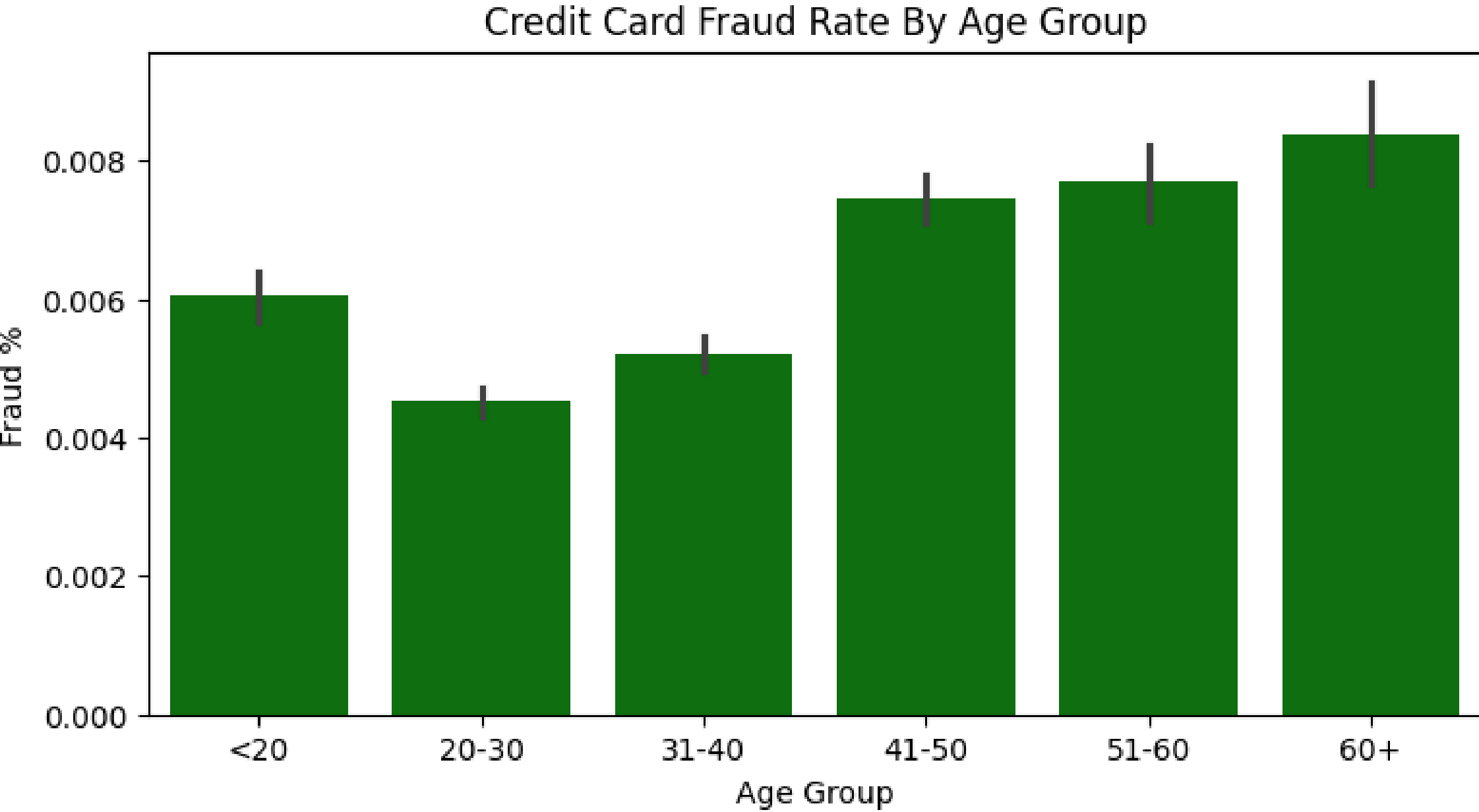
We chosen a dataset from Kaggle that was generated as we were not able to obtain live data to use due to GDPR (General Data Protection Regulation).

Certain pre-existing features from the dataset were used to generate new features that would best fit our model (age, month, day, etc.)

After obtaining those features, it was then standardized and scaled.



Are certain age groups more likely to be fraud victims?

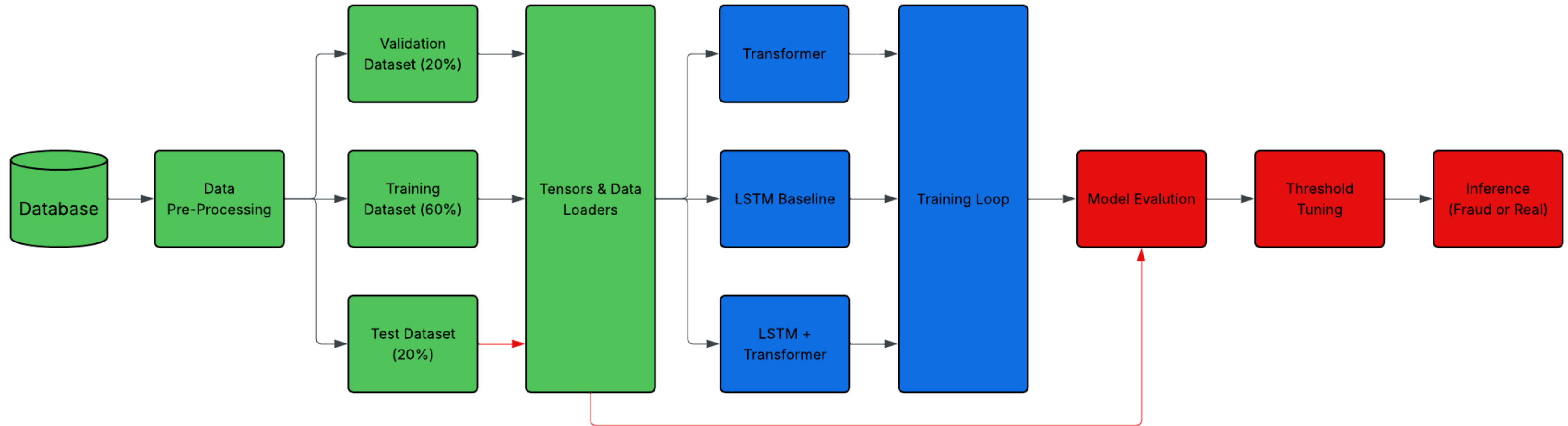


Deep Learning Methods

- Traditional ML models are not complex enough
- Sequential data
- Previous work leverages models such as CNNs
- Combining Deep Learning architectures

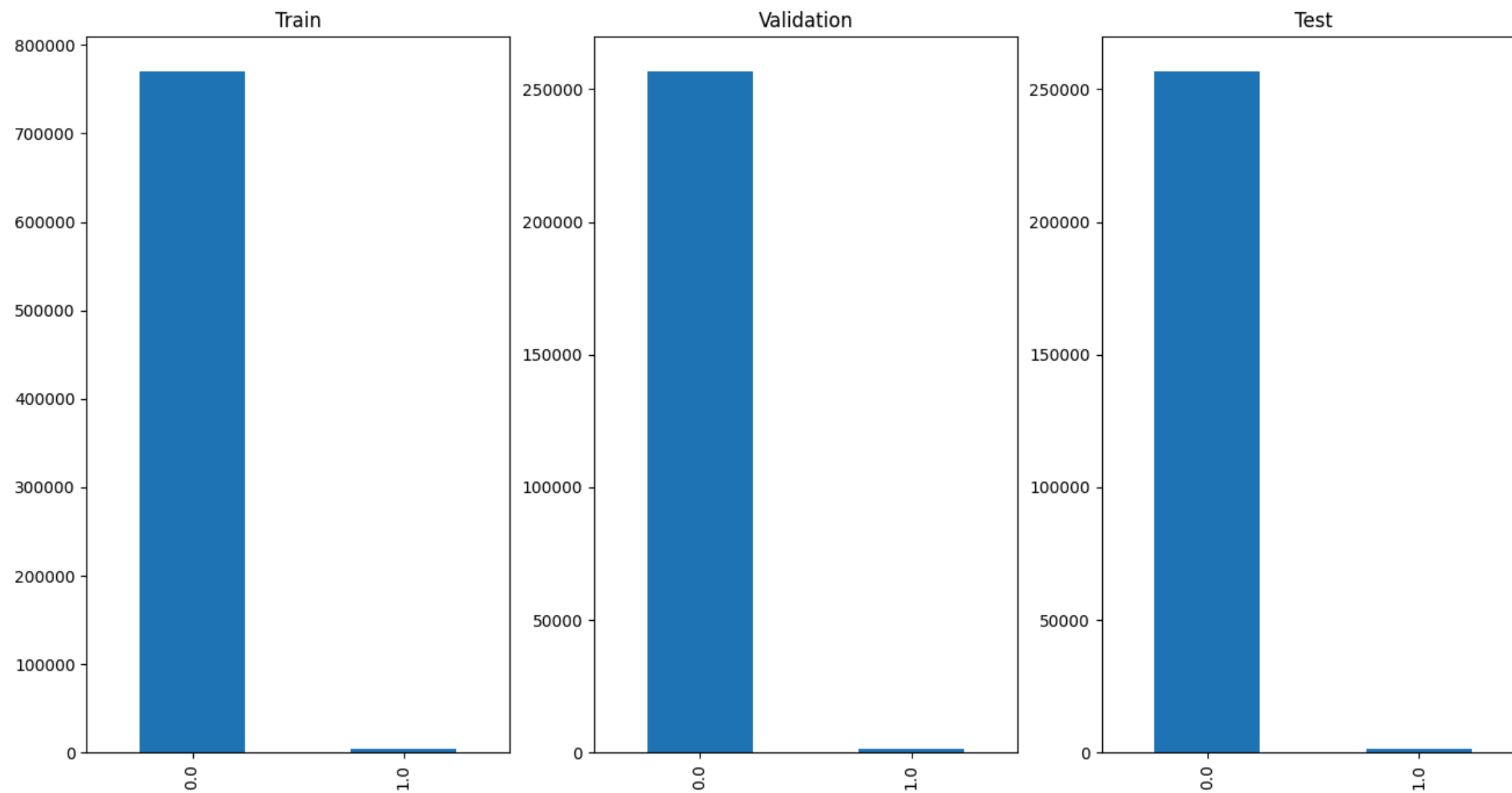


Model Architecture





Model Training



- LSTM: great at dealing with sequential data
- Transformer: better than LSTM at pattern recognition due to attention mechanisms
- LSTM + Transformer:
 - leveraging the temporal strength of LSTM + pattern recognition of a transformer
- Binary Cross Entropy
- Focus on reducing the influence of class imbalance
 - Adam(): reduce errors
 - Scheduler: alter step size
 - Patience(2): cycles to wait before giving up
 - pos_weight: punishing model
 - encouraging to end at early epochs
- Epochs: 25
 - stopped at 17, 7, 11





Evaluation

Model	ROC-AUC	AVG Precision	F1	RECALL	PRECISION
LSTM	0.953	0.746	0.58	0.83	0.45
TSF	0.943	0.677	0.82	0.82	0.42
LSTM TSF	0.945	0.749	0.73	0.78	0.68

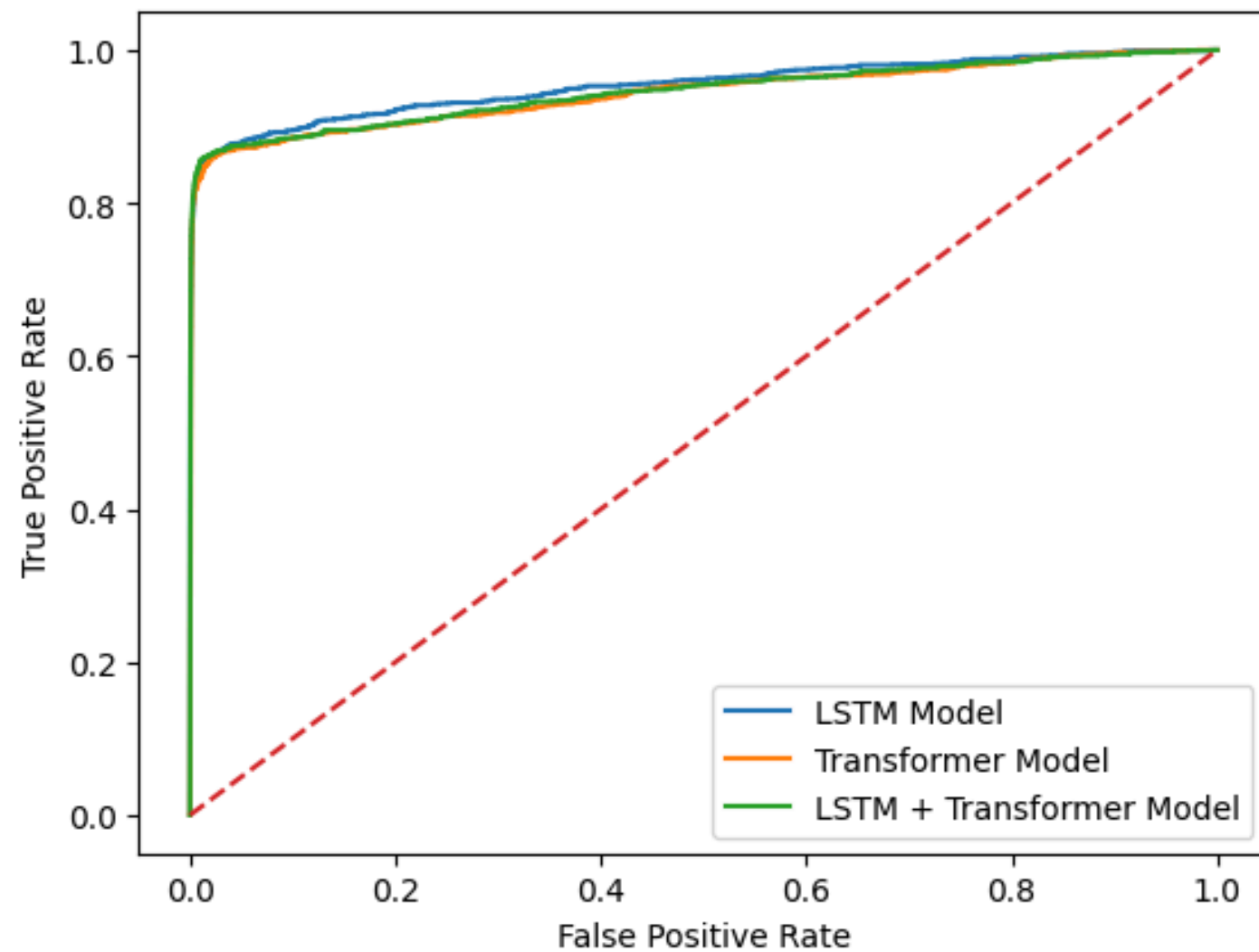
- F1 score went up from 58% to 73%
 - LSTM TSF is correct 73% of the time
- Decrease in false predictions = increase in precision (more accurate)
- Fusion model is overall better for real-life predictions





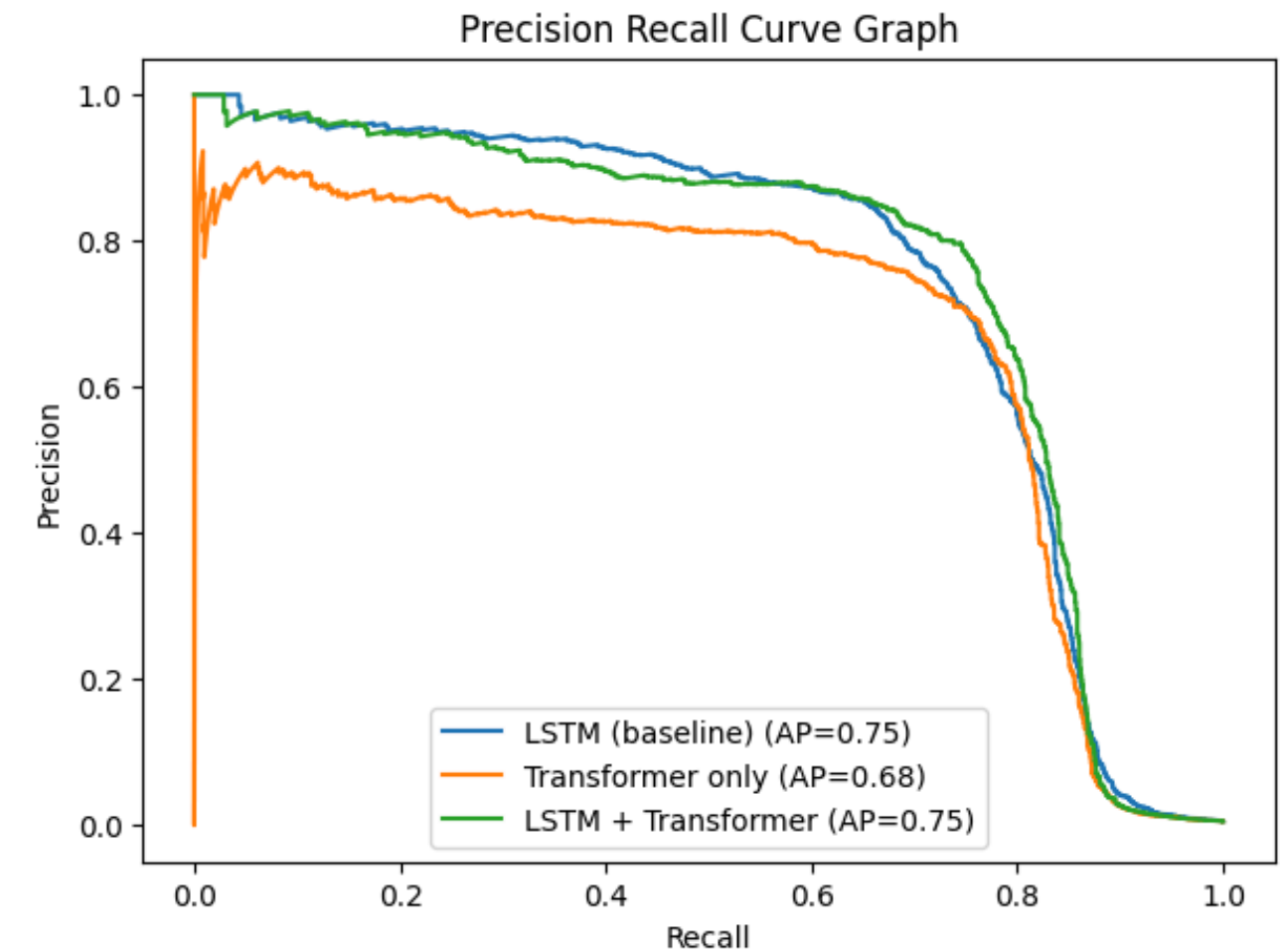
Evaluation

ROC Curves



- good at identifying fraud, the red line is random guessing
- The closeness reflects the scores being close
- not helpful as much because of the imbalance

Precision Recall Curves



- As mentioned, I can see that TSF is doing worse than others
- Individuals perform okay, but our stacked model is **better**
- good at catching ~ 80% of fraud





Next Steps

- Build on our current model
 - CNN / RNN
- More fraudulent datasets
- Add more attention heads
- Better Feature Engineering
- Better technique in balancing data (SMOTE)
- Comparing against statistical approaches



References

1. Sudjianto, Agus, Sheela Nair, Ming Yuan, Aijun Zhang, Daniel Kern, and Fernando Cela-Díaz. 2010. "Statistical Methods for Fighting Financial Crimes." *Technometrics* 52 (1): 5–19. doi:10.1198/TECH.2010.07032.
2. Khalid, Abdul Rehman, Nsikak Owoh, Omair Uthmani, Moses Ashawa, Jude Osamor, and John Adejoh. 2024. "Enhancing Credit Card Fraud Detection: An Ensemble Machine Learning Approach" *Big Data and Cognitive Computing* 8, no. 1: 6. <https://doi.org/10.3390/bdcc8010006>
3. Bin Sulaiman, Rejwan, Vitaly Schetinin, and Paul Sant. "Review of Machine Learning Approach On Credit Card Fraud Detection" *Human-Centric Intelligent Systems* 2, no. 1 (2022): 55-68.
4. Dhankhad, Sahil, Emad Mohammed, and Behrouz Far. "Supervised Machine Learning Algorithms For Credit Card Fraudulent Transaction Detection: A Comparative Study" In 2018 IEEE international conference on information reuse and integration (IRI), pp. 122-125. IEEE, 2018.
5. Saputra, Adi. "Fraud Detection Using Machine Learning in E-commerce." *International Journal of Advanced Computer Science and Applications* 10, 332–339, no. 9 (2019).
6. Esenogho, Ebenezer, Ibomoiye Domor Mienye, Theo G. Swart, Kehinde Aruleba, and George Obaido. "A Neural Network Ensemble With Feature Engineering For Improved Credit Card Fraud Detection" *IEEE access* 10 (2022): 16400-16407.
7. Roseline, J. Femila, G. B. S. R. Naidu, V. Samuthira Pandi, S. Alamelu alias Rajasree, and N. Mageswari. "Autonomous Credit Card Fraud Detection Using Machine Learning Approach" *Computers and Electrical Engineering* 102 (2022): 108132.
8. Shenoy, Kartik and Harris, Brandon 2020 "Credit Card Transactions Fraud Detection Dataset." Kaggle. <https://www.kaggle.com/datasets/kartik2112/fraud-detection>.



Thank You!

